Do

What if

How

What

Where

Actionable, Closed-Loop World Representations

L4 Unified Embodied Modeling for Navigation & Action

- VLM
- VLN
- VLA
- World Models
- Unified Embodied Foundation Models

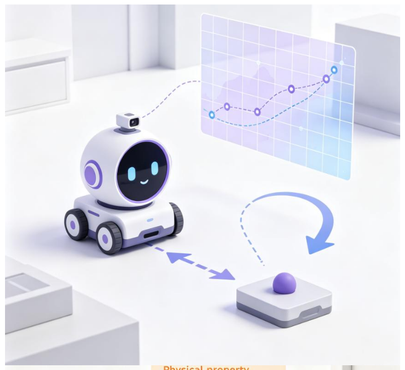
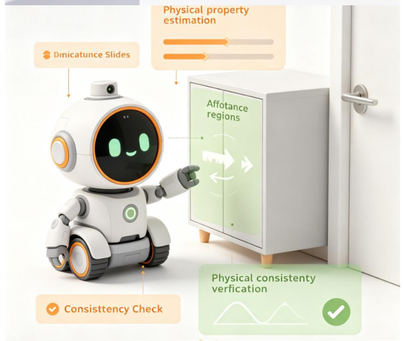
L3 Physical & Functional Understanding {G, S, F, P, C}

- Physical Property Estimation
- Affordance Reasoning
- Causal Dynamics & Prediction
- Physical Consistency Verification

L2 Semantic Understanding {G, S}

- Object-Level Semantics
- Relational Structure
- Language-Aligned Open Semantics
- Semantic-Geometry Fusion

L1 Geometric Reconstruction {G}

- Offline Reconstruction
- Feed-Forward Geometry
- Online SLAM & Mapping

Data & Simulation Substrate